Easy Problems

1. **Don't drink and derive.**

$$\frac{df}{dx} = \lim_{\Delta x \to 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

2. **Sigma sigma boy.**

$$\sum_{b=6}^{10} \sum_{o=7}^{11} \sum_{y=8}^{12} boy$$

3. **iff.**

$$p \leftrightarrow q \equiv (p \to q) \land (q \to p)$$

4. **Baye's Ball.**

$$P(A \cap B_j) = P(A)P(B_j \mid A)$$

5. **Saan.**

$$D_{\vec{u}} f = \nabla f \cdot \vec{u}$$

6. **Chevron James.**

$$\overrightarrow{Lebron} = \langle j, a, m, e, s \rangle$$

7. **Smile ka muna!**

$$\overset{\cdot\cdot}{\smile}$$

8. **The floor is lava.**

$$\lfloor x \rfloor = a^2 lv$$

Average Problems

1. **Identitad.**[1]

$$I_n = \begin{bmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \end{bmatrix}$$

2. **Boom Boom Bakudan.**

$$\sum_{k=0}^{n} (-1)^k \binom{n}{k} \binom{n+k}{m} = (-1)^n \binom{n}{m-n}$$

3. **Fibonacci.**

$$F_n = F_{n-1} + F_{n-2} = \frac{\phi^n - \psi^n}{\phi - \psi} = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}$$

[1] **Hint**: Adjust the `column-gap` of `math.mat` to `1em`.

4. **Sophomore's Dream**.

$$\int_0^1 x^{-x}\,dx = \sum_{n=1}^{\infty} n^{-n}$$

5. **Box lang**.

$$\begin{aligned} H &= \begin{bmatrix} \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \\ -\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} & \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 1 & 1 & 1 \\ -1 & 1 & -1 & 1 \\ -1 & -1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{bmatrix} \end{aligned}$$

6. **Karla Karla Estrada Estrada**.[2]

$$\mathsf{tugon}(i) = \begin{cases} \text{Karla} & \text{if } i \in \{0,1\} \\ \text{Estrada} & \text{if } i \in \{2,3\} \end{cases}$$

7. **Divide and Conquer**.

$$T(n) = \Theta\left(n^p\left(1 + \int_1^n \frac{f(x)}{x^{p+1}}\,dx\right)\right)$$

8. **Fibonacci Part II**.

$$\begin{aligned} \mathscr{F}(z) &= \frac{z}{1-z-z^2} \\ &= \frac{z}{(1-\phi z)\left(1-\hat{\phi} z\right)} \\ &= \frac{1}{\sqrt{5}}\left(\frac{1}{1-\phi z} - \frac{1}{1-\hat{\phi} z}\right) \end{aligned}$$

Difficult Problems

1. **We love LaTeX**[3]

$$\LaTeX$$

2. **Nahuhulog**.

$$\begin{aligned} f(\mathbf{x}\text{-avg}) - f(\mathbf{x}^*) &= f\left(\frac{\sum_{t=0}^{T-1} \mathbf{x}^{(t)}}{T}\right) - f(\mathbf{x}^*) \\ &\le \frac{\sum_{t=0}^{T-1} f\left(\boldsymbol{x}^{(t)}\right)}{T} - f(\mathbf{x}^*) \\ &\le B + \frac{\Phi(0)}{T} \end{aligned}$$

[2] **Hint**: The font of tugon is sans.

[3] ~~You really don't need to force yourself to do this~~